UNIT 8 PRACTICAL GEOMETRY

8.1 CONSTRUCTION OF QUADRILATERALS

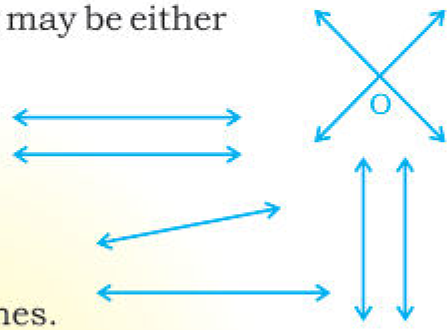
8.1.1 Define and depict two converging (non-parallel) lines and find the angle between them without producing the lines

When we draw two lines in a plane, they may be either

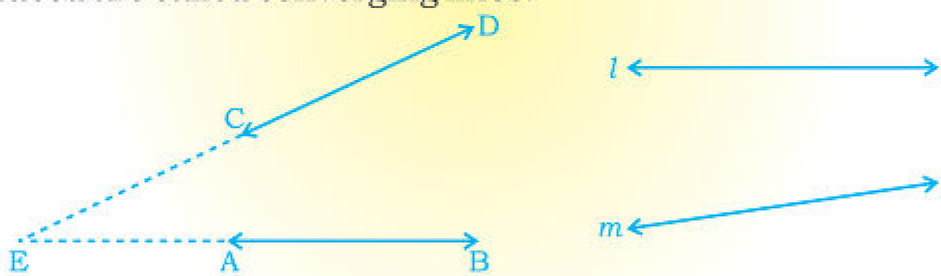
(i) intersecting lines at a point

or (ii) parallel lines

or (iii) neither intersecting nor parallel but in such a state that they will intersect when produced. Such two lines are called converging lines.



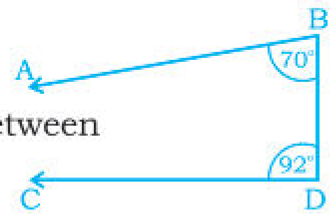
Definition: Two lines or line segments in a plane which are neither intersecting nor parallel but they will intersect at a point when produced are called converging lines.



In the above figure $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ are converging lines because when $\overline{DC}$ and $\overline{BA}$ are produced they will meet at a point (say E) making an angle ($\angle AEC$). This phenomenon has been shown by dotted lines.

Similarly l and m will intersect when produced on the converging side. It should be noted that in converging lines one side is converging (getting closer and closer) whereas the other side is non-converging or diverging (i.e. getting farther and farther away). In the lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$, the side with points A and C is converging whereas the side with points B and D is diverging (or non-converging).

To find the measure of an angle between two converging lines without producing them:

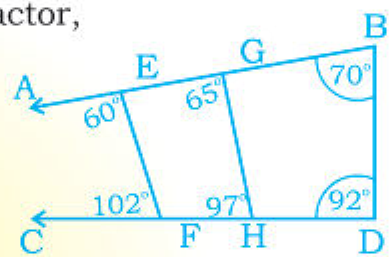


Given: Two converging lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$

Required: To find the measure of the angle between $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ without producing them.

Steps of construction:

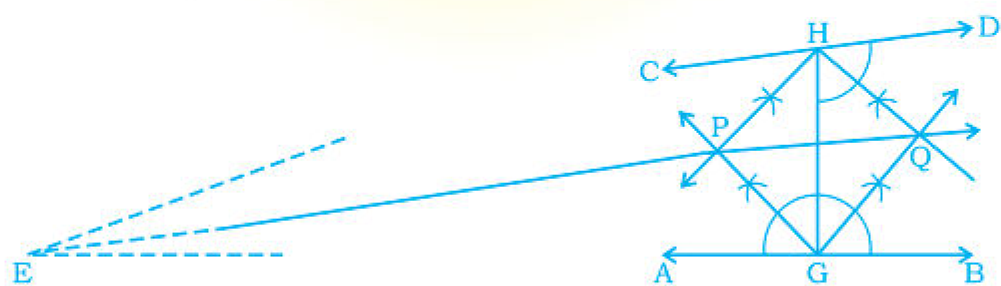
1. Join B to D (Points on the diverging side)
2. Measure $\angle ABD$ with the help of a protractor, we have $m\angle ABD = 70^\circ$
3. Measuring $\angle CDB$ with a protractor, we have $m\angle CDB = 92^\circ$
4. Therefore, the measure of angle (say x) between the converging lines is:



$$m\angle x = 180^\circ - (m\angle ABD + m\angle CDB)$$
$$= 180^\circ - (70^\circ + 92^\circ)$$
$$= 180^\circ - 162^\circ = 18^\circ$$

Note that in whatever way, we join the points on the diverging side of two lines, sum of measures of both the angles will be the same as shown by $\overline{EF}$ and $\overline{GH}$.

8.1.2 Bisect the angle between two converging lines without producing them



Given: Two converging lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$

Required: To bisect the angle (that would form on producing the converging sides of lines) without producing them.

Steps of construction:

1. At a convenient place, draw a transvarsal intersecting $\overleftrightarrow{AB}$ at G and $\overleftrightarrow{CD}$ at H.
2. Draw the bisectors of $\angle AGH$ and $\angle CHG$ intersecting each other at point P.
3. Draw the bisectors of $\angle BGH$ and $\angle DHG$ intersecting each other at point Q.
4. Draw $\overline{QP}$ and produce it to any point E.

$\overleftrightarrow{QP}$ is the required bisector of the angle between two converging lines without producing them.

8.1.3 Construct a square when:

(a) Measure of its diagonal is given.

(b) Difference between the measures of a diagonal and a side is given.

(c) Sum of the measures of a diagonal and a side are given.

(a) Construct a square when measures of its diagonal is given

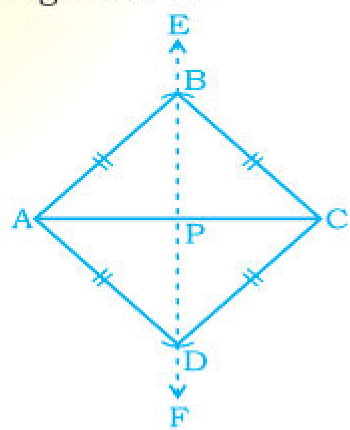
Example 1. Draw square ABCD whose diagonal measures 6cm.

Given: In square ABCD, $m\overline{AC} = 6$ cm

Required: To construct a square ABCD from the given data

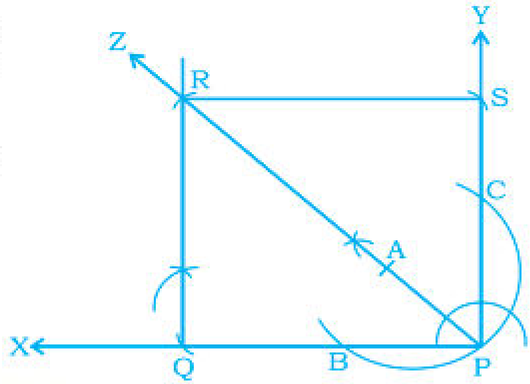
Steps of construction:

1. Draw $\overline{AC}$, such that $m\overline{AC} = 6$cm
2. Draw $\overleftrightarrow{EF}$, the right bisector of $\overline{AC}$ intersecting it at point P.
3. Take P as centre and with radius $m\overline{PA}$ (or $m\overline{PC}$) draw arcs on both sides of $\overline{AC}$ cutting $\overleftrightarrow{EF}$ at B and D.
4. Draw $\overline{AB}$, $\overline{BC}$, $\overline{CD}$ and $\overline{DA}$.
   Now ABCD is the required square.



(b) Construct a square when difference between the measures of a diagonal and a side is given.

Example 2. Construct square PQRS when the difference between its diagonal and a side is 3 cm.



Given: In Square PQRS, difference between its diagonal and one side is 3 cm

Required: To construct square PQRS from the given data

Steps of construction:

1. Draw a ray $\overrightarrow{PX}$.
2. At point P draw $\overrightarrow{PY} \perp \overrightarrow{PX}$.
3. Draw $\overrightarrow{PZ}$, the bisector of $\angle XPY$
4. From $\overrightarrow{PZ}$, cut off $\overrightarrow{PA}$ such that $m\overline{PA} = 3$ cm
5. Taking point A as centre and with radius 3cm, draw an arc cutting $\overrightarrow{PX}$ at B and PY at point C.
6. Taking points B and C as cnetres and with the same radius (i.e., 3cm) draw arcs cutting $\overrightarrow{PX}$ at Q and $\overrightarrow{PY}$ at point S.
7. Taking points Q and S as centres and with radius m $\overline{PQ}$ (or $m\overline{PS}$) draw arcs cutting one another and $\overrightarrow{PZ}$ at point R.
8. Draw $\overline{QR}$ and $\overline{SR}$. Now PQRS is the required square.

Note: In diagonal $\overline{PR}$, $m\overline{AR} = m\overline{PQ}$ (or $\overline{PS}$ etc.) side of square and $m\overline{AP} = 3$cm (difference between diagonal and side)

[After the construction is complete, verify the lengths of the diagonal and the side].

(c) Construct a square when the sum of measures of its diagonal and side is given.

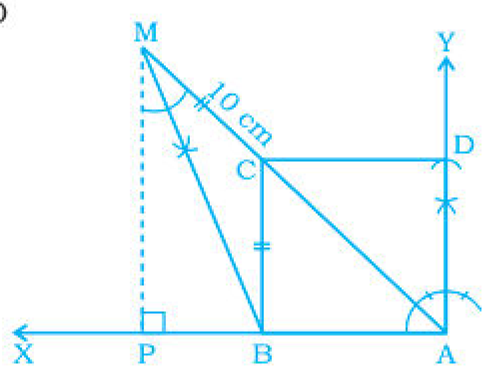
Example. Construct square ABCD, when the sum of its diagonal and one side is 10 cm.

Given: In square ABCD, the sum of its diagonal and a side is 10 cm.

Required: To construct square ABCD from the given data.

Steps of construction:

1. Draw a ray $\overrightarrow{AX}$
2. Draw $\overrightarrow{AY} \perp \overrightarrow{AX}$
3. Draw $\overline{AM}$ the bisector of $\angle XAY$ such that $m\overline{AM} = 10$ cm
4. Draw $\overline{MP} \perp \overrightarrow{AX}$
5. Draw $\overline{MB}$, the bisector of $\angle AMP$ meeting $\overrightarrow{AP}$ at point B.



6. Through point B, draw $\overline{BC} \parallel \overline{MP}$ meeting $\overline{AM}$ at point C.
7. From $\overline{AY}$, cut off $\overline{AD} \cong \overline{BC}$
8. Draw $\overline{CD}$.

Now quadrilateral ABCD is the required square.

Note: Sum of a diagonal and a side of square = 10 cm, it has been divided into two parts
(i) diagonal $\overline{AC}$ and (ii) $m\overline{MC} = m\overline{BC}$ (side of square)

8.1.4 Construct a Rectangle

(a) When two sides are given

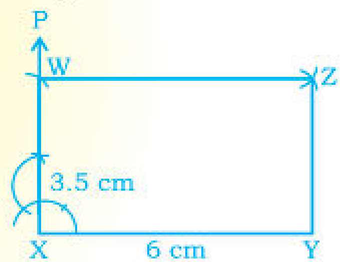
Example 1.

Construct rectangle XYZW in which $m\overline{XY} = 6$ cm and $m\overline{YZ} = 3.5$ cm

Given: $m\overline{XY} = 6$ cm, $m\overline{YZ} = 3.5$ cm, are sides of a rectangle XYZW.

Required: To construct rectangle XYZW from the given data.

Steps of construction:



1. Draw $\overline{XY}$, 6 cm long
2. At point X, draw $\overrightarrow{XP} \perp \overline{XY}$
3. Take point X as centre and with radius 3.5 cm, draw an arc to cut $\overrightarrow{XP}$ at W.
4. Take point W as centre and with radius $m\overline{XY}$ draw an arc.
5. Take point Y as centre and with radius $m\overline{XW}$ draw an arc which cuts the previous arc at Z.
6. Draw $\overline{WZ}$ and $\overline{YZ}$.

Now XYZW is the required rectangle.

(b) When the diagonal and a side are given.

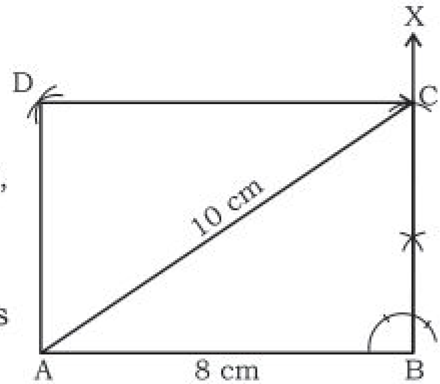
Example 2.

Construct rectangle ABCD in which diagonal m $\overline{AC} = 10$ cm and a side $\overline{AB} = 8$ cm.

Given: In rectangle ABCD $m\overline{AC} = 10$ cm and $m\overline{AB} = 8$ cm.

Required: To construct a rectangle ABCD from the given data.

Steps of construction



1. Draw $\overline{AB}$, 8cm long
2. At point B, draw $\overrightarrow{BX} \perp \overline{AB}$
3. Take point A as centre, with radius 10 cm, draw an arc which cuts $\overrightarrow{BX}$ at C.
4. Join point A to C (Draw $\overline{AC}$).
5. Take point A as centre, with radius m $\overline{BC}$, draw an arc.
6. Take point C as centre, with radius m $\overline{AB}$, draw another arc which cuts the previous arc at point D.
7. Draw $\overline{AD}$ and $\overline{CD}$.

Now ABCD is the required rectangle.

8.1.5 Construct a Rhombus

(a) When one side and the base angle are given.

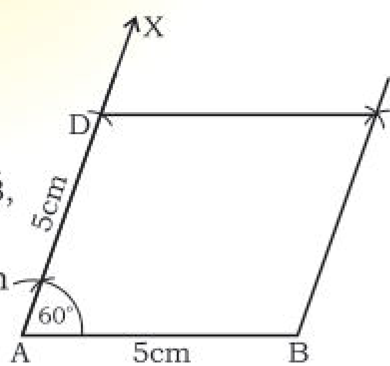
Example 1.

Construct Rhombus ABCD whose sides are 5 cm each and measure of base angle is 60°.

Given: In Rhombus ABCD, $m\,\overline{AB} = 5$ cm, $m\,\angle A = 60^\circ$.

Required: To construct Rhombus ABCD from the given data.

Steps of construction



1. Draw $\overline{AB}$, 5 cm long
2. At point A, construct $\angle BAX$ such that $m\,\angle BAX = 60^\circ$
3. Taking point A as centre, with radius $m\,\overline{AB}$, draw an arc to intersect $\overrightarrow{AX}$ at point D
4. Taking points B and D as centres, with the same radius of $m\overline{AB}$, draw arcs cutting each other at C.
5. Draw $\overline{BC}$ and $\overline{DC}$.

Now Quad. ABCD is the required Rhombus.

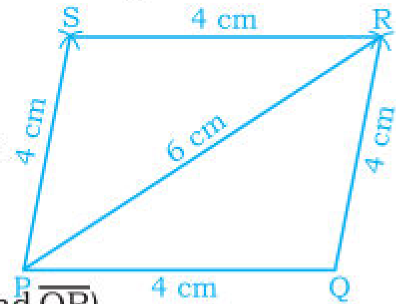
(b) When one side and a diagonal are given.

Example 2. Construct Rhombus PQRS when $m\overline{PQ} = 4$ cm and $m\overline{PR} = 6$ cm.

Given: In Rhombus PQRS, $m\overline{PQ} = 4$ cm, $m\overline{PR} = 6$ cm.

Required: To construct Rhombus PQRS from the given data.

Steps of construction:



1. Draw $\overline{PQ}$, 4 cm long.
2. Take point P as centre, with radius 6 cm draw an arc.
3. Take point Q as centre, with radius $m\overline{PQ}$, draw another arc to cut the previous arc at point R.
4. Join point P to R and Q to R. (Draw $\overline{PR}$ and $\overline{QR}$)
5. Taking points P and R as centres and with radius $m\overline{PQ}$. draw arcs to cut each other at point S.
6. Join point S to P and to R. (Draw $\overline{PS}$ and $\overline{PR}$)

Now Quadrilateral PQRS is the required Rhombus.

8.1.6 Construct a Parallelogram

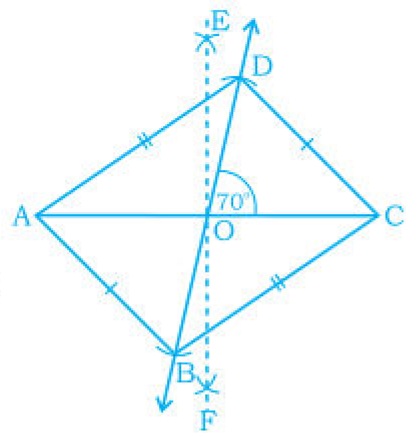
(a) When the two diagonals and angle between them are given.

Example 1. Construct a $||^m$ ABCD, in which $m\overline{AC} = 6$ cm, $m\overline{BD} = 4.5$ cm and angle between them is of 70°.

Given: In $||^m$ ABCD, $m\overline{AC} = 6$ cm , $m\overline{BD} = 4.5$ cm and $m\angle COD = 70^\circ$

Required: To construct $||^m$ ABCD from the given date.

Steps of construction



1. Draw $\overline{AC}$, 6 cm long.
2. Draw $\overleftrightarrow{EF}$ the right bisector of $\overline{AC}$ intersecting $\overline{AC}$ at point O.
3. At point O, with the help of protractor construct $\angle COD$ of measure 70°
4. Take point O as centre and with radius $\frac{4.5}{2} = 2.25$ cm, draw arcs on both sides to cut $\overrightarrow{OD}$ at D and B.

5. Draw $\overline{AB}$, $\overline{AD}$, $\overline{BC}$ and $\overline{CD}$

Now Quad. ABCD is the required parallelogram.

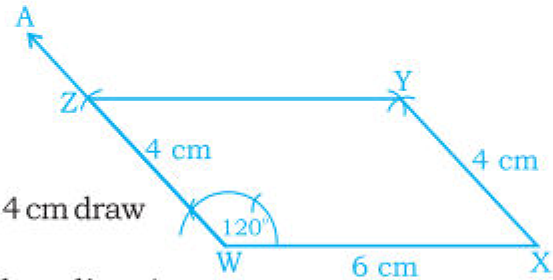
(b) When two adjacent sides and the angle included between them is given.

Example 2. Construct $\|^{m}$ WXYZ in which $m\,\overline{WX} = 6$ cm, $m\,\overline{WZ} = 4$ cm and $m\angle XWZ = 120^\circ$.

Given: In $\|^{m}$ WXYZ, $m\,\overline{WX} = 6$ cm, $m\,\overline{WZ} = 4$ cm and $m\angle XWZ = 120^\circ$

Required: To construct $\|^{m}$ WXYZ from the given data.

Steps of construction:



1. Draw $\overline{WX}$, 6 cm long.
2. At point W, construct $\angle XWA$ such that $m\angle XWA = 120^\circ$.
3. Take point W as centre, with radius 4 cm draw an arc to cut $\overrightarrow{WA}$ at Z.
4. Take point X as centre and with radius 4 cm draw an arc.
5. Take point Z as centre and with radius 6 cm draw another arc to cut the previous arc at Y.

Now Quad WXYZ is the required parallelogram.

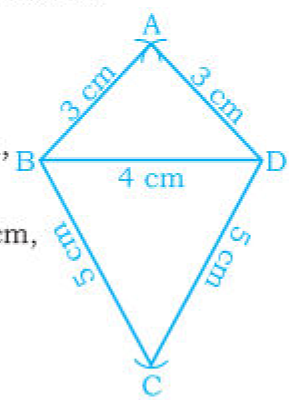
8.1.7 Construct a Kite when two unequal sides and a diagonal are given.

Example 3. Construct Kite ABCD in which $m\,\overline{AB} = 3$ cm, $m\,\overline{BC} = 5$ cm and diagonal $\overline{BD} = 4$ cm

Given: In Kite ABCD, $m\,\overline{AB} = 3$ cm, $m\,\overline{BC} = 5$ cm and $m\,\overline{BD} = 4$ cm

Required: To construct Kite ABCD from the given data.

Steps of construction:



1. Draw $\overline{BD}$, 4 cm long.
2. Take pointB and D as centres and with radius 3 cm, draw arcs to cut one another at point A.
3. Again take points B and D as centres and with radius 5 cm, draw arcs to cut one another at point C.
4. Draw $\overline{AB}$, $\overline{AD}$, $\overline{BC}$ and $\overline{DC}$

Now Quad. ABCD is the required Kite.

8.1.8 Construct a Regular Pentagon when measurement of a side is given.

Example 4. Construct a Regular Pentagon ABCDE, each of whose side measures 4 cm.

Given: In Regular Pentagon ABCDE, $m\overline{AB} = 4$ cm

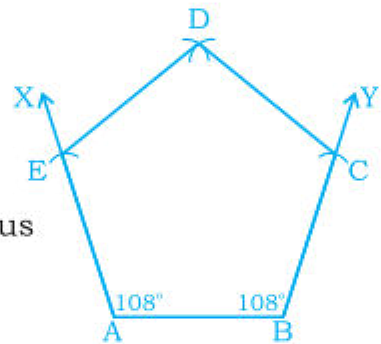
$$m\angle A = m\angle B = \left(\frac{2(5)-4}{5}\right) rt\angle s \quad \text{[from unit-7]}$$

$$= \left(\frac{10-4}{5}\right) \times 90^\circ \qquad = \frac{6}{\cancel{5}} \times \cancel{90}^{18} = 108^\circ$$

Required: To construct Regular Pentagon ABCDE from the given data.

Steps of construction:

1. Draw $\overline{AB}$, 4 cm long.
2. At points A and B, construct, with the help of protractor, $\angle BAX$ and $\angle ABY$ such that each angle measures 108°.
3. Take points A and B as centres, and with radius $m\overline{AB}$, draw arcs to cut $\overrightarrow{AX}$ at E and $\overrightarrow{BY}$ at point C.



4. Take points C and E as centres and with the same radius draw arcs to cut one another at point D.
5. Draw $\overline{CD}$ and $\overline{ED}$.

Now polygon ABCDE is the required Regular Pentagon.

8.1.9 Construct a Regular Hexagon when a side is given.

Example 5.

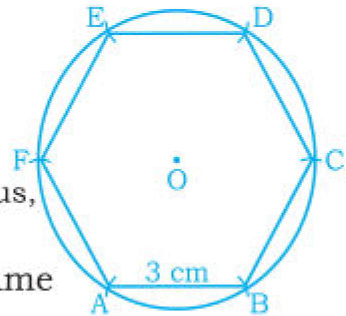
Construct Regular Hexagon ABCDEF each of whose side measures 3 cm.

Given: In Regular Hexagon, $m\overline{AB} = 3$ cm

Required: To construct Regular Hexagon ABCDEF from the given data.

Steps of construction:

1. At a convenient place, take point O as centre and with radius 3 cm construct a circle
2. Take a point A on the circle.
3. Take point A as centre, with radius 3 cm, draw an arc to cut the circle at B.
4. Then taking point B as centre, with the same radius, draw another arc to cut the circle at C.
5. Again take point C as centre and with the same radius, draw an arc to cut the circle at point D and repeat the same process with Centres D and E to cut the circle at points E and F respectively.
6. Draw $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, $\overline{DE}$, $\overline{EF}$ and $\overline{FA}$.



Now polygon ABCDEF is the required Regular Hexagon.

EXERCISE 8.1

1. Draw similar converging lines in your exercise book as given below, then:

(i) Draw the bisector of the angle formed by the converging lines and (ii) Find the measure of the angle without producing the lines.

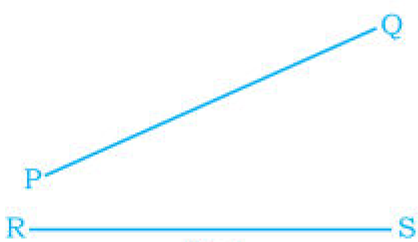


Fig. 1

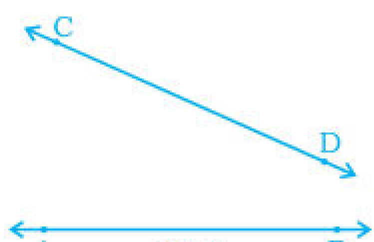


Fig. 2

2. (a) Construct square ABCD in which measure of its diagonal is as under:

(i) $m\overline{AC} = 6$ cm (ii) $m\overline{BD} = 7$ cm (iii) $m\overline{AC} = 8$ cm

(b) Construct square PQRS when the difference between its diagonal and a side is:

(i) 2.5 cm (ii) 3 cm (iii) 3.5 cm (iv) 4.5 cm

(c) Construct square WXYZ when the sum of its diagonal and a side is:

(i) 10 cm (ii) 12 cm (iii) 11 cm (iv) 8 cm

3. (a) Construct rectangle PQRS when its two sides are:

(i) $m\overline{PQ} = 6$ cm, $m\overline{QR} = 4$ cm (ii) $m\overline{PQ} = 4$ cm, $m\overline{QR} = 7$ cm

(b) Construct rectangle ABCD, when its diagonal and one side are given:

(i) $m\overline{AC} = 6$ cm, $m\overline{AB} = 4$ cm (ii) $m\overline{BD} = 7$ cm, $m\overline{AB} = 5$ cm

4. (a) Construct Rhombus ABCD when one side and base angle are given:

(i) $m\overline{AB} = 4$ cm, $m\angle A = 120^\circ$ (ii) $m\overline{AB} = 5$ cm, $m\angle A = 60^\circ$

(b) Construct Rhombus PQRS, when one side and a diagonal are given:

(i) $m\overline{PQ} = 5$ cm, m $\overline{QS} = 4$ cm (ii) $m\overline{PQ} = 4.5$ cm, $m\overline{PR} = 7$ cm

5.(a) Construct $||^m$ PQRS, when two diagonals and angle between them are given:

(i) $m\overline{QR} = 8$ cm, $m\overline{QS} = 6$ cm, $m\angle QOR = 60^\circ$

(ii) $m\overline{PR} = 6$ cm, $m\overline{QS} = 8$ cm, $m\angle QOR = 120^\circ$

(b) Construct $||^m$ ABCD, when two adjacent sides and included angle are given:

(i) $m\overline{AB} = 6$ cm, $m\overline{BC} = 4$ cm, $m\angle B = 70^\circ$

(ii) $m\overline{AB} = 4$ cm, $m\overline{AD} = 5$ cm, $m\angle A = 100^\circ$

6. Construct Kite ABCD, when two unequal sides and a diagonal are given:

(i) $m\overline{AB} = 4$ cm, $m\overline{BC} = 7$ cm, $m\overline{AC} = 9$ cm

(ii) $m\overline{AB} = 4$ cm, $m\overline{BC} = 6$ cm, $m\overline{BD} = 4$ cm

7. Construct Regular Pentagon ABCDE each of whose side is:

(i) 3.5 cm (ii) 4 cm (iii) 4.5 cm (iv) 5 cm

8. Construct Regular Hexagon ABCDEF each of whose side is:

(i) 3 cm (ii) 3.5 cm (iii) 2.5 cm (iv) 4 cm

8.2 CONSTRUCTION OF A RIGHT ANGLED TRIANGLE

(a) Construct a right angled triangle when hypotenuse and one side are given:

Example 1. Construct right-angled-Triangle ABC, in which hypotenuse $m\overline{BC} = 10$ cm and $m\overline{AB} = 6$ cm

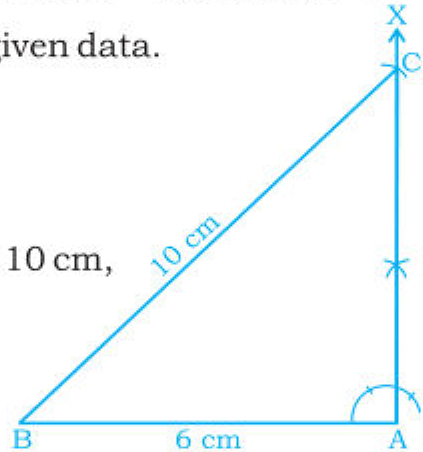
Given: In right angled ΔABC, $m\overline{BC} = 10$ cm, $m\overline{AB} = 6$ cm, $m\angle A = 90^\circ$

Required: To construct ΔABC from the given data.

Steps of construction:

1. Draw $\overline{AB}$, 6cm long.
2. At point A, draw $\overrightarrow{AX} \perp \overline{AB}$.
3. Take point B as centre and with radius 10 cm, draw an arc which cuts $\overrightarrow{AX}$ at point C.
4. Join points B and C.(Draw $\overline{BC}$)

Now ΔABC is the required right triangle.



(b) Construct a right angled triangle when the hypotenuse and the Vertical Height from the vertex to the Hypotenuse are given.

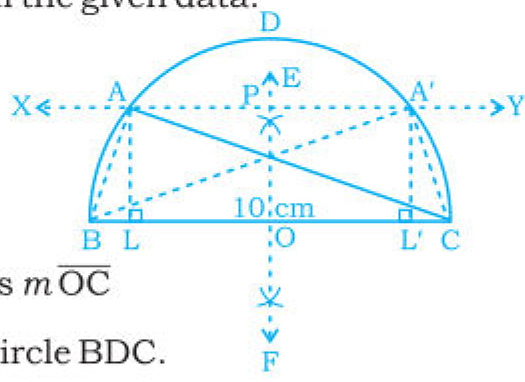
Example 2. Construct a right-angled ΔABC, in which hypotenuse $\overline{BC}$ is 10 cm and vertical height from point A to $\overline{BC}$ is 3.8 cm.

Given: In right angled ΔABC, $m\angle A = 90^\circ$, $m\overline{BC} = 10$ cm, $m\overline{AL} = 3.8$ cm ($\overline{AL} \perp \overline{BC}$)

Required: To construct ΔABC from the given data.

Steps of construction:

1. Draw $\overline{BC}$, 10 cm long.
2. Draw $\overleftrightarrow{EF}$ the right bisector of $\overline{BC}$ intersecting it at O.



3. Take O as centre, and with radius $m\overline{OC}$ or $m\overline{OB}\left(\frac{10}{2} = 5 \text{ cm}\right)$ draw semi-circle BDC.

4. Take a point P on $\overrightarrow{OE}$ such that $m\overline{OP} = 3.8$cm

5. Through point P, draw $\overleftrightarrow{XY} \parallel \overline{BC}$ (i.e., draw $\overleftrightarrow{XY} \perp \overline{OP}$) cutting the circle at points A and A′.

6. Draw $\overline{AB}$ and $\overline{AC}$ (Also draw $\overline{A'B}$ and $\overline{A'C}$)

Now ΔABC (and $\Delta A'BC$) are the required rt. angled triangles in which $m\overline{AL} = m\overline{A'L'} = 3.8$ cm

Note: In any semi-circle, inscribed triangle is a right angled triangle.

EXERCISE 8.2

1. Construct the following right angled triangles when:

(a) Hypotenuse = 10 cm and measure of one side = 8 cm

(b) Hypotenuse = 13 cm and measure of one side = 12 cm

(c) One side $\overline{PQ}$ = 5 cm, one side $\overline{SQ}$ = 6 cm

2. Construct the following right angled triangles when:

(a) Hypotenuse = 10 cm and vertical height from vertex to Hypotenuse = 3 cm

(b) Hypotenuse = 12 cm and vertical height from vertex to Hypotenuse = 4 cm

(c) Vertical height $\overline{PM}$ = 3 cm, Vertical height $\overline{SN}$ = 3.5 cm

(d) Vertical height $\overline{PM}$ = 2.5 cm, Vertical height $\overline{SN}$ = 3 cm

3. Construct the right angled triangles ABC and DBC on either sides of Hypotenuse $\overline{BC}$, 10 cm long when:

(a) Vertical height $\overline{AL}$ = 3.5 cm, $m\overline{DC}$ = 5 cm

(b) Vertical height $\overline{AL}$ = 3 cm, $m\overline{DB}$ = 6 cm

REVIEW EXERCISE 8

1. Four options are given for each statement. Tick the one which is correct.

(i) Measure of the angle formed by two converging lines (without producing them) is equal to:

(a) Difference of measures of two angles formed by joining the end points of diverging side.

(b) Difference of 180° and the sum of the measures of two angles formed by joining the end-points of diverging side.

(c) 180° – (sum of the measures of angles formed by drawing any transversal any where cutting the two lines).

(d) None of these.

(ii) A polygon with sum of the measures of all interior angles equal to four right angles is ____________.

(a) Triangle (b) Quadrilateral (c) Pentagon (d) Hexagon

(iii) A polygon with sum of the measures of all interior angles equal to two right angles is ____________.

(a) Triangle (b) Quadrilateral (c) Pentagon (d) Hexagon

(iv) A polygon with sum of measures of all interior angles equal to eight right angles is ____________.

(a) Triangle (b) Quadrilateral (c) Pentagon (d) Hexagon

(v) A polygon with sum of measures of all interior angles equal to six right angles is ____________.

(a) Triangle (b) Quadrilateral (c) Pentagon (d) Hexagon

(vi) In a rectangle the diagonals __________________________.

(a) bisect each other at right angles (b) are perpendicular to each other
(c) intersect each other at mid-point (d) None of these

(vii) Square is a ______________________.

(a) Rectangle (b) Regular Polygon
(c) Quadrilateral having unequal sides (d) None of these

(viii) Measure of each interior angle of a Regular Hexagon is ____________.

(a) 108° (b) 120° (c) 140° (d) None of these

2. Construct the following:

(i) Square PQRS such that $m\overline{QS} = 6$ cm

(ii) Square WXYZ when the difference of its diagonal and side is 2.5 cm

(iii) Square ABCD when the sum of its diagonal and side is 12 cm.

(iv) Rectangle ABCD, when $m\overline{AB} = 4$ cm, $m\overline{BD} = 6$ cm

(v) Rhombus PQRS, when $m\overline{PQ} = 5$ cm, $m\angle P = 110^\circ$

(vi) Rhombus ABCD, when m $\overline{BC} = 5$ cm, $m\overline{BD} = 7$ cm

(vii) $//^m$ WXYZ whose diagonals are 6 cm and 9 cm and the angle between them is of 70°.

(viii) Kite ABCD when $m\overline{AB} = 6$ cm, $m\overline{BC} = 3$ cm, $m\overline{BD} = 4$ cm.

(ix) Regular Pentagon PQRST when $m\overline{PQ} = 4$ cm.

(x) Regular Hexagon ABCDEF when $m\overline{AB} = 3.5$ cm

3. Draw any two converging line segments $\overline{AB}$ and $\overline{CD}$ and draw the bisector of angle formed by these without producing these lines.

4. Draw any two converging line segments $\overline{PQ}$ and $\overline{PS}$ converging toward points Q and S.

(a) Find the angle between them (to be formed).

(b) Draw the bisector of the angle to be formed without producing the lines.

SUMMARY

- Converging lines are non-parallel lines (or line segments) which will intersect if produced.
- Angle between the converging lines without producing them is equal to 180° – (sum of the measures of two angles formed by joining the non-converging end points).
- By joining the In-centre and ex-centre of the triangle (that would be formed if lines are produced) we get the bisector of the angle by converging lines (without producing them) because the bisector of angles are concurrent with these points.